



JARDEM

AI-Powered Hand Rehabilitation

Every session measured. Every recovery counted.

Healthcare Innovation · 2026



Stroke leaves millions without access to proper rehabilitation

Stroke is the **third leading cause of disability worldwide**. Recovery requires months of daily, repetitive hand exercises — but for most patients in rural and low-resource areas, that care is out of reach.



No Therapist Nearby

Most stroke survivors in rural areas have no access to a qualified rehabilitation specialist.

Without guidance, patients don't know if they are exercising correctly.



Costly Clinic Visits

Transport costs, long distances, and session fees make regular clinic attendance impossible for most patients.

Each missed visit is lost recovery time.



No Clinical Record

Doctors have no way of knowing what patients actually do at home between appointments.

Recovery is guessed, not measured.

Existing solutions don't solve this

Current devices are:

- **Expensive** — imported hardware
- **Hard to repair** — imported parts
- **Clinically blind** — no data to doctors

Existing apps require:

- Therapist online, stable internet, digital literacy



The result is a clear gap: there is no tool that bridges home rehabilitation and clinical oversight — one that makes home sessions as measurable and accountable as clinic visits, without requiring the patient to travel or the doctor to be online.

Jardem — rehabilitation that watches, corrects, and records



Real-Time Movement Correction

The camera watches your hand during every exercise and tells you instantly if you're doing it right.

The same feedback a physiotherapist gives, available at home, any time.



Automatic Clinical Reports

After every session the app records range of motion, rep count, and movement quality.

Every four weeks it compiles a one-page clinical report for the doctor.



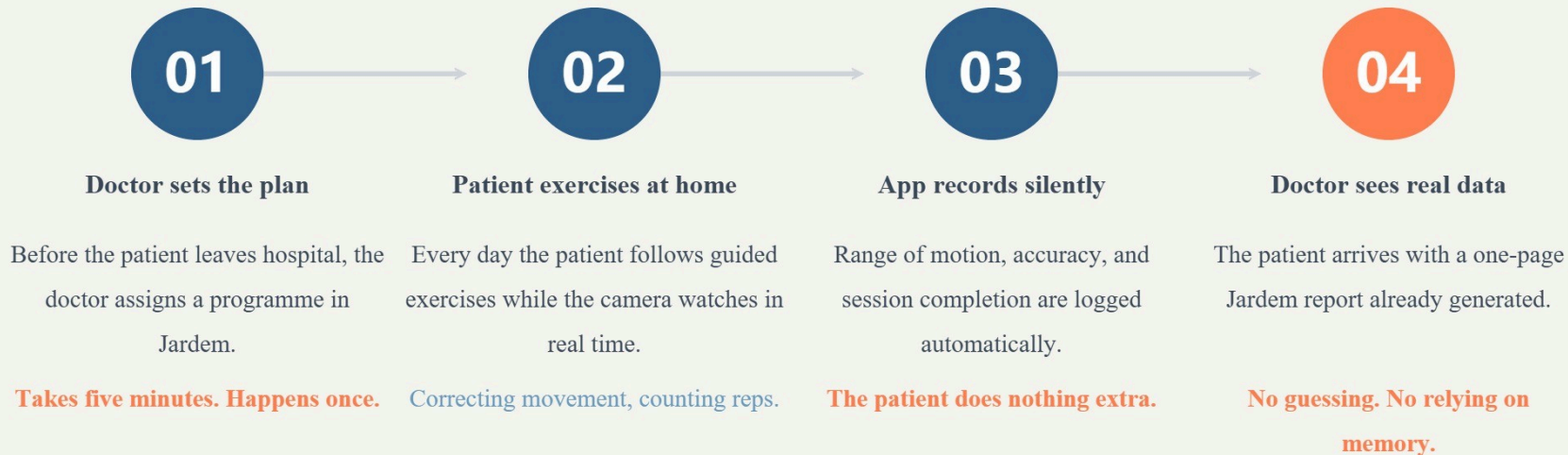
Instant Safety Escalation

If the patient reports pain or the camera detects dangerous movement — the app immediately stops and displays the doctor's phone number.

No waiting. No delay.

No wearables. No special equipment. Just a smartphone.

Four steps. No clinic needed.



From hospital discharge to follow-up — **every session tracked, every movement measured**

What makes this possible

Real-Time Hand Tracking

MediaPipe detects **21 hand landmarks** from the phone camera — tracking finger angles, range of motion, and speed. **No wearables required.**

Automatic Clinical Logging

Every session is recorded into a structured clinical note. ROM, rep count, movement quality, and pain flags compile into a report for doctor review.

On-Device Processing

All video analysis happens **locally**. No video is sent to a server. **Privacy protected by design** — works fully offline.



Built for vulnerable patients — with no compromises



Auto-Stop

If pain is reported twice, the exercise is immediately paused.

Won't resume until the doctor reactivates it.



Instant Escalation

When something goes wrong, the app stops immediately.

Displays the doctor's phone number for instant calling.



Doctor in Control

The doctor sets the plan. The AI delivers within safe boundaries only.

No clinical decision without doctor approval.



Privacy Protected

No video is stored or transmitted. All processing is local. Patient data never leaves the device.

Privacy by design, not by promise.



Clear Messaging

Every screen includes a clear disclaimer — Jardim is a support tool, not a replacement for clinical care.

Transparency builds trust with patients and doctors.

The need is urgent and the timing is right

258.4 → 433.7

Cerebrovascular disease per 100,000 in Kazakhstan

2015 to 2020 — *Zhakhina et al., 2022*

80%+

of stroke-related disability in LMICs

Jardem is designed for this reality:



Low-Data — works on minimal internet connection



Works Offline — daily sessions require no connectivity at all



Language-Flexible — available in any language



Any Smartphone — works on devices patients already own



No New Doctor Workflow — reports go to appointments that were already happening. Jardem fits into existing care pathways without adding burden.

Where we are and where we're going

Where We Are Now

Working on prototype with:

- Camera-based hand tracking
- Real-time movement feedback
- Guided exercise sessions
- Automatic session logging
- Pain and difficulty flagging



What Comes Next

01

Clinical Report Validation

Partner with physiotherapists to confirm automatic reports match manual assessments.

02

Pilot Study

Supervised pilot with real post-stroke patients at discharge and follow-up.

03

Expansion

Scale across Kazakhstan and open-source for global low-resource deployment.

Every session measured. Every recovery counted.

Millions of stroke survivors go home with a sheet of paper and no way to know if they are recovering correctly.

Jardem changes that. A camera that watches and corrects. An app that records every session. A report that gives doctors real data. And an instant escalation system for when something goes wrong.

J A R D E M

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